V.L Memorial Public School

475 Vijay Nagar

CHAPTER-11-QUESTIONS-CLASS6TH-SCIENCE

Class 06 - Science (Curiosity)

Section A

1. Why does the Pole Star help locate the North direction? **[1]**

   a) It changes position
   b) It appears brightest
   c) It forms three stars
   d) It appears stationary

2. Which of the following is not a planet of the Sun? **[1]**

   a) Jupiter
   b) Pluto
   c) Neptune
   d) Saturn

3. Which star is not visible from the southern hemisphere? **[1]**

   a) Sirius
   b) Betelgeuse
   c) Canopus
   d) Pole Star

4. The stars forming a recognizable shape is called **[1]**

   a) system
   b) constellation
   c) asteroids
   d) galaxy

5. What does the term nakshatra denote in Indian astronomy? **[1]**

   a) A comet or meteor
   b) A galaxy or cluster
   c) A star or group
   d) A planet or orbit

6. How many constellations were officially listed by the International Astronomical Union (IAU)? **[1]**

   a) 52
   b) 103
   c) 88
   d) 75

7. Which star pattern is commonly represented as a hunter? **[1]**

   a) Ursa Major constellation
   b) Taurus constellation
   c) Orion constellation
   d) Canis Major constellation

8. Which of the following does not belong to the family of solar system? **[1]**

   a) Comet
   b) Planet
   c) Meteors
   d) Galaxy

9. What is excessive artificial light at night called? **[1]**

   a) Light pollution
   b) Sound pollution
   c) Air pollution
   d) Water pollution

10. How long does it take for the Moon to complete one revolution around the Earth? **[1]**

a) 365 days
b) 27 days
c) 30 days
d) 24 hours

11. Why would someone prefer a moonless night for observing the Pole Star? **[1]**

a) The Pole Star gets overshadowed by the moon's brightness.
b) For traditional beliefs associated with the Pole Star.
c) A moonless night prevents weather changes.
d) The Pole Star only appears during new moons.

12. Which small objects revolve between the orbits of Mars and Jupiter? **[1]**

a) Asteroids
b) Meteorites
c) Comets
d) Satellites

13. The Halley's Comet is seen after every **[1]**

a) 76 months
b) 56 years
c) 56 months
d) 76 years

14. Why were imaginary lines drawn between stars in constellation figures? **[1]**

a) For showing actual boundaries
b) For changing star positions
c) For easier identification
d) For measuring star temperatures

15. Where is the night sky best viewed? **[1]**

a) Tall building terraces
b) Open dark areas
c) Crowded market places
d) Busy city streets

16. Which unit is used to measure astronomical distances? **[1]**

a) Light year
b) Leap year
c) Century
d) Sound year

17. What are constellations still commonly defined as, despite modern astronomy's preference? **[1]**

a) Unique star clusters
b) Groups of stars forming patterns
c) Planets orbiting stars
d) Dim stars with no patterns

18. **Assertion (A):** The Sun is the center of the universe. **[1]**
Reason (R): The universe consists of many galaxies beyond our solar system.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

19. **Assertion (A):** Stars appear to rise and set in the sky. **[1]**
Reason (R): This is because the Earth rotates on its axis.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.
d) A is false but R is true.

20. **Assertion (A):** The Sun plays a crucial role in sustaining life on Earth. **[1]**
Reason (R): The energy produced by the Sun is the primary source of light and heat for the planets within our Solar System.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

21. **Assertion (A):** In urban areas, it is challenging to view many stars at night. **[1]**
Reason (R): Light pollution, caused by excessive artificial lighting, obstructs the view of stars in urban settings.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

22. **Assertion (A):** The Moon has phases due to its position relative to the Earth and Sun. **[1]**
Reason (R): The Moon emits its own light.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

23. **Assertion (A):** All constellations are visible from every place on Earth. **[1]**
Reason (R): Some constellations are visible only from certain hemispheres.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

24. **Assertion (A):** Light pollution increases our ability to see more stars. **[1]**
Reason (R): Light pollution causes the sky to appear brighter and blocks fainter stars.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

25. What is the role of the International Astronomical Union in defining constellations? **[1]**
26. What unit is approximately equal to the distance between the Sun and Earth? **[1]**
27. Define what a constellation is and how it is identified in the sky. **[1]**
28. Explain why scientists study galaxies beyond the Milky Way Galaxy. **[1]**
29. Why can more stars generally be seen from villages than from big cities? **[1]**
30. Name two sky mapping apps useful for identifying stars, constellations, and planets. **[1]**
31. Why is the Pole Star not visible from the southern hemisphere of Earth? **[1]**

Section B

32. Match the following: **[2]**

Column (A)	Column (B)
(a) Light pollution	(i) Helps in finding directions
(b) Dark sky reserves	(ii) Makes it difficult to see stars
(c) Magnetic compass	(iii) Helps identify stars and constellations
(d) Sky mapping apps	(iv) Areas where light pollution is controlled

33. Match the following: **[2]**

Column (A)	Column (B)
(a) Orion	(i) Known as Saptaṛiṣhi in India
(b) Canis Major	(ii) Contains the brightest star, Sirius
(c) Ursa Major	(iii) Represented as a hunter in the night sky
(d) Pole Star (Polaris)	(iv) Appears stationary and helps locate the North direction

34. How does the presence of tall buildings and trees affect night sky viewing, and what can be done to mitigate these effects during observations? **[2]**
35. What technological tools assist in exploring objects not visible to the naked eye in space, and how do they function? **[2]**
36. Explain the cultural significance of constellations in navigation throughout history. **[2]**
37. Imagine creating a new educational interactive exhibit about the solar system. What innovative features would you add to engage students and enhance learning? **[2]**
38. Discuss the differences between inner and outer planets in terms of composition and size. **[2]**
39. Examine the reasons why the inner planets appear smaller when compared to outer planets. **[2]**
40. If you were to design a mission to discover life on exoplanets, what key factors would you need to consider? **[2]**

Section C

41. Why do planets appear different from stars when observed in the night sky? **[3]**
42. How do the stories associated with the Big Dipper and the Little Dipper reflect cultural diversity in star interpretation? **[3]**
43. Discuss the significance of dark sky reserves and how they aid in night sky observations. **[3]**
44. Explain the role of the International Astronomical Union (IAU) in defining modern constellation boundaries and the significance of this standardization. **[3]**
45. How can someone living in a big city improve their night sky observations despite prevalent light pollution? **[3]**
46. Evaluate how technological resources, like sky mapping apps, have impacted amateur astronomy. **[3]**
47. How could knowledge of star patterns help a traveller when modern navigation technology is unavailable? **[3]**

Section D

48. **Read the following text carefully and answer the questions that follow:** **[4]**

There are many galaxies in outer space beyond the Milky Way Galaxy. Scientists study them to understand the stars, galaxies, and the universe. We do not know yet if life exists elsewhere in the universe. The search for life has been directed mostly towards exoplanets — the planets discovered revolving around other stars in our galaxy. Till now, scientists have not found any evidence of life, but this search is on and continuing.

Questions:

a. Why are exoplanets important for scientists in their search for life beyond Earth? **(1)**

b. Which star helps travellers find direction in the Northern Hemisphere, and how? **(1)**

c. How can studying galaxies beyond the Milky Way help humanity in the future? **(2)**

OR

Why do scientists continue searching for life on other planets even though no evidence has been found yet? **(2)**

49. **Read the following text carefully and answer the questions that follow:** **[4]**

If it is a clear cloudless night, a large number of stars may be visible in the sky. If you stay in a big city, you may find that the sky is rarely clear and only a few stars are seen in the night sky. This is due to light pollution, smoke and dust. The presence of excessive artificial light at night time is referred to as light pollution. In villages or areas where there is less light pollution, a larger number of stars can be seen. Light pollution is growing sharply globally. This is reducing our ability to enjoy and study objects in the night sky. Some dark sky reserves and parks have been established around the world. Light pollution in the reserves is controlled to preserve dark skies for research. There are a few organisations which are working to educate people about reducing light pollution.

a. What is the term used to describe excessive artificial light at night? (1)

b. Why are fewer stars visible in the night sky in big cities compared to villages? (1)

c. Explain how light pollution affects our ability to observe the night sky. (2)

OR

What are dark sky reserves, and why are they important? (2)

50. **Read the following text carefully and answer the questions that follow:** **[4]**

A planet is a large, nearly spherical object that revolves around the Sun. Our Earth is a planet as it revolves around the Sun. The Earth takes nearly one year to complete one revolution. Like the Earth, there are other planets that revolve around the Sun. While revolving around the Sun, the Earth is also rotating about its axis. For one full rotation, the Earth takes about 24 hours which is called a day. Like the Earth, other planets also rotate about their axes while revolving around the Sun. The eight planets, in order of their increasing distance from the Sun, are Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune. The inner four planets nearest to

the Sun-Mercury, Venus, Earth and Mars-are smaller in size. They have solid surfaces with rocks on them.

a. How many planets are there in our solar system, and what are the inner four planets called? (1)
b. What is the time taken by the Earth to complete one rotation about its axis? (1)
c. Explain the difference between rotation and revolution of the Earth. (2)

OR

Why are the inner four planets (Mercury, Venus, Earth, and Mars) called terrestrial planets? (2)

Section E

51. Evaluate how city conditions differ from a dark open area for night sky observation. **[5]**
52. How can sky maps, constellation images, and a compass help identify a constellation? **[5]**
53. How did Chandrayaan-3 contribute to India's lunar exploration? **[5]**
54. Why do craters remain visible on the Moon's surface for a very long time? **[5]**
55. Discuss the impact of cultural storytelling on the perception and classification of constellations across different societies. **[5]**
56. What role do constellations play in preserving cultural heritage, and how can they be further leveraged to educate new generations about astronomy and history? **[5]**
57. Compare the major star patterns and names associated with Orion, the Big Dipper, the Little Dipper, and the Pole Star. **[5]**

Section F

58. **State whether the given statement is True or False:** **[9]**
    (a) Sky Map and Stellarium are apps designed to help identify stars, constellations, and planets. **[1]**
    (b) In big cities, a larger number of stars can be seen due to clearer skies. **[1]**
    (c) The Earth's natural satellite is called Mercury. **[1]**
    (d) Earth is our home planet. **[1]**
    (e) Sirius is part of the constellation Gemini. **[1]**
    (f) The Milky Way is the name of our solar system. **[1]**
    (g) Dark sky reserves are areas where light pollution is minimized to preserve dark skies for scientific study. **[1]**
    (h) The Earth is a natural satellite of the Moon. **[1]**
    (i) Pole star is a member of the solar system. **[1]**
59. **Fill in the blanks:** **[9]**
    (a) The Sun, planets, their moons, along with asteroids and comets, form the _____. **[1]**
    (b) A _____ is known as the Red Planet due to its reddish soil. **[1]**
    (c) _____ is the third nearest planet to the sun. **[1]**
    (d) The Moon is Earth's natural _____. **[1]**
    (e) A/An _____ is made of metals and rocky material. **[1]**
    (f) The inner planets have a _____ surface. **[1]**
    (g) The weather in Nubra is almost _____. **[1]**

(h) _____ is called the windy planet because of the continuous strong winds that blow on it. **[1]**

(i) A large group of stars and other celestial bodies is called a _____. **[1]**